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ASSIGNMENT REPORT

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Comparative Analysis of Unconstrained and State-Constrained Optimal Vaccination Strategies in an SIR Epidemic Framework

This report presents a comparative investigation into the application of optimal control to a susceptible-infected-recovered (SIR) epidemic model, based on the reference journal by Della Rossa et al. (2025). Vaccination $v(t)$ is the control variable, deployed to minimise the combined cost of intervention effort and infected burden under two paradigms: an unconstrained environment (Case I) and a system governed by a strict Intensive Care Unit (ICU) capacity limit on the infected state (Case II).

The simulation yielded an objective function value of $J_{\text{Case I}} = 22.60$ for the unconstrained scenario and a higher $J_{\text{Case II}} = 22.81$ for the state-constrained framework, an increase of $\Delta J \approx 0.21$ (about 1%). Introducing the path constraint $i(t) \leq i_{\text{M}} = 0.04$ restricts the feasible state-space region available to the solver. Forced to hold the infected trajectory below the ICU ceiling, the optimiser cannot follow its natural, mathematically ideal path; total efficiency drops and the overall societal cost rises. The increase therefore quantifies the price of keeping infections within hospital capacity.

Investigating the trajectories reveals distinct behaviour. In Case I, the infected state I rises unhindered and overshoots the ICU level: it crosses i_{M} at $t \approx 14.9$, peaks at $I_{\max} \approx 0.0412$ at $t \approx 17.9$, then falls back below i_{M} at $t \approx 21.1$. The control follows the classic bang-bang structure set by the switching function $\varphi = \lambda_{\text{v}} - p_{\text{s}} \cdot s$, vaccinating at the maximum rate v_{M} and switching once to zero at $t * \approx 29.9$. This overshoot above ICU capacity is precisely what motivates the constraint imposed in Case II.

In Case II, the constraint clamps the infection peak to exactly $i_{\mathrm{M}} = 0.04$. Since the susceptible state is strictly decreasing, the infected state cannot remain on the boundary over any interval; it touches i_{M} at a single contact point $t_0 \approx 17.9$, not along a boundary arc, in agreement with the paper's central theorem. At this instant the co-state p_{i} shows a downward jump, the signature of the active state-constraint measure multiplier, visible as the sharp drop in the Case II co-state graph. The theoretical optimum remains bang-bang with at most one switch; the control returned by IPOPT declines gradually after the contact rather than switching sharply to zero, a discretisation artefact rather than the exact theoretical control. These results validate the state-constrained findings of the reference journal.

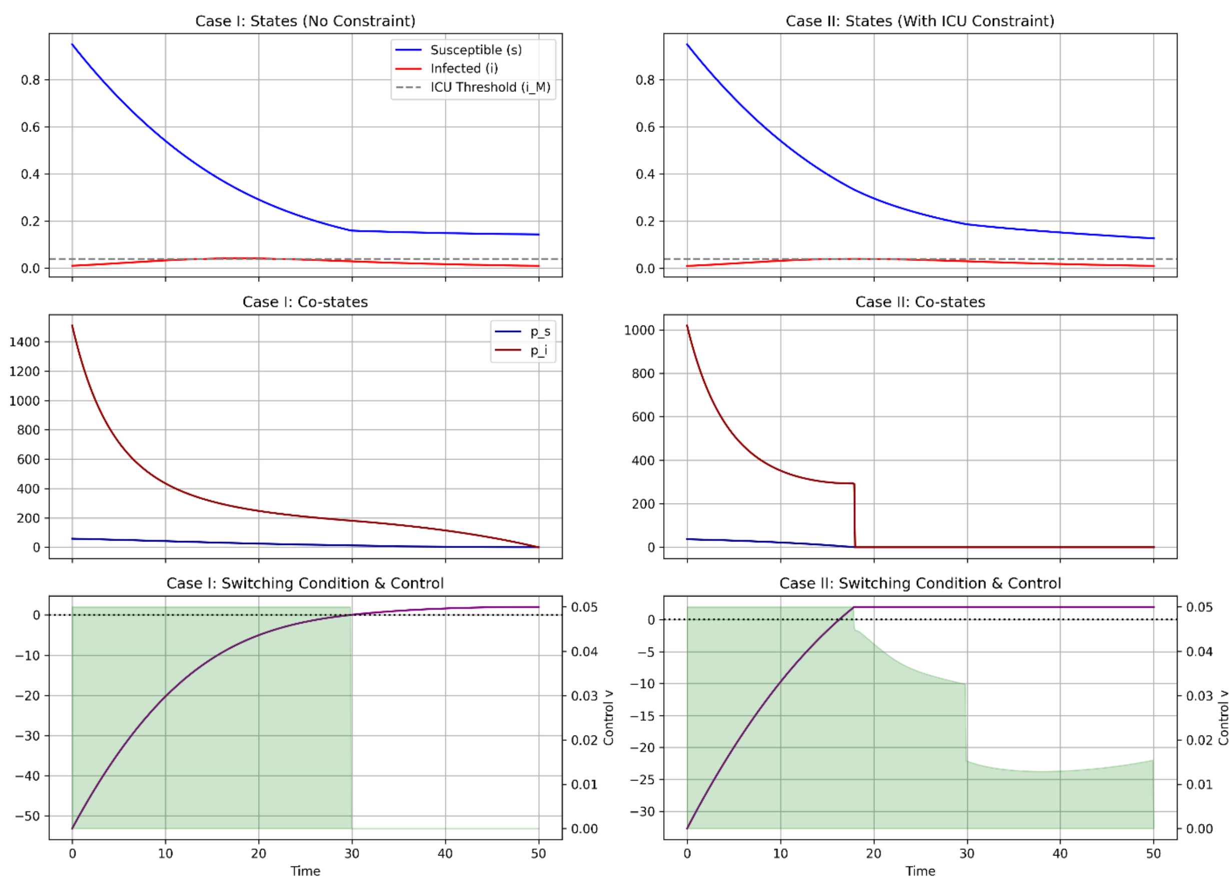

Figure 1: Comparative profiles of State Trajectories (S, I), Co-States (p_s, p_i), Switching Condition φ, and Optimal Vaccination Controls (V) for Case I (Unconstrained) and Case II (State-Constrained).

References

Della Rossa, M., Freddi, L., & Goreac, D. (2025). Optimality of Vaccination for an SIR Epidemic with an ICU Constraint. *Journal of Optimization Theory and Applications*, *204*(1). https://doi.org/10.1007/s10957-024-02598-w